MAWLANA BHASHANI SCIENCE AND TECHNOLOGY UNIVERSITY

SANTOSH, TANGAIL-1902



DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Lab Report

Lab Report No : 01

Lab Report on : Image Reading, Display, Grayscale Conversion, and Normalization using MATLAB

Course Title : Digital Image Processing Lab

Course Code : ICT-4104

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Date of Performance: 28/04/26 Date of Submission: 05/05/26

Introduction:

Digital Image Processing (DIP) involves analyzing and manipulating images using computer algorithms. Images consist of pixels that store color or intensity information. In this experiment, a color image is converted to grayscale and its pixel values are normalized using MATLAB, which are basic and essential steps in image processing.

Objective:

- To read a color image in MATLAB
- To display the image
- To convert the color image into grayscale
- To normalize pixel values for further processing

Software Used:

- MATLAB R2018b

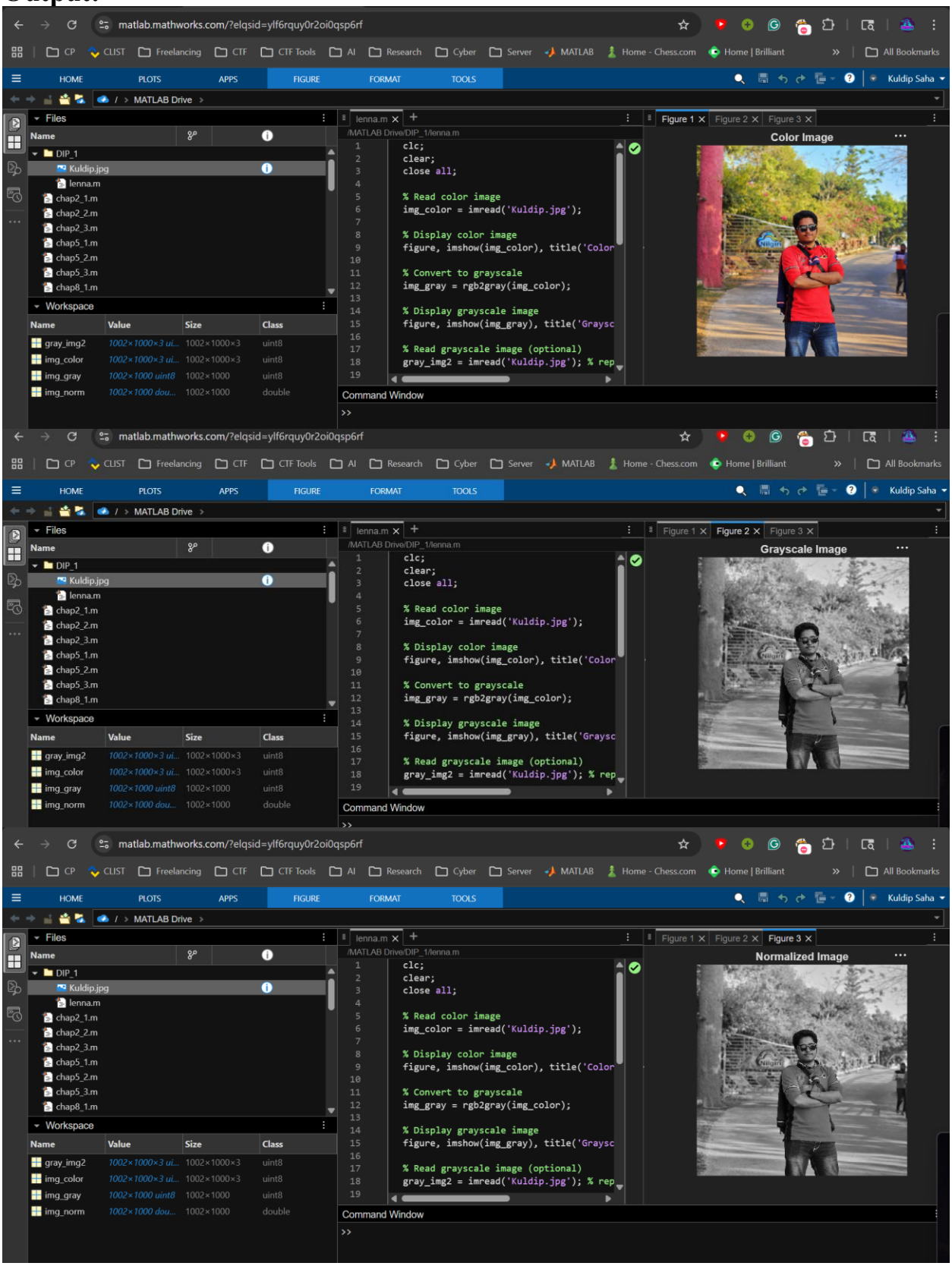
Theory:

Digital images are composed of pixels, each representing intensity values. In color images, each pixel consists of three channels: Red, Green, and Blue (RGB).
Grayscale conversion reduces the image from three channels to one by combining RGB values using weighted averages.
Normalization is the process of scaling pixel values (usually from 0–255) into a range of 0–1. This is useful for many image processing and machine learning tasks.

MATLAB Code:

```
clc;
clear;
close all;
% Read color image
img_color = imread('Kuldip.jpg');
% Display color image
figure, imshow(img_color), title('Color Image');
% Convert to grayscale
img_gray = rgb2gray(img_color);
% Display grayscale image
figure, imshow(img_gray), title('Grayscale Image');
% Read grayscale image (optional)
gray_img2 = imread('Kuldip.jpg'); % replace with grayscale file
if needed
% Normalize pixel values
img_norm = double(img_gray) / 255;
% Display normalized image
figure, imshow(img_norm), title('Normalized Image');
```

Output:


matlab.mathworks.com/?elqsid=ylf6rquy0r2oi0qsp6rf
HOME PLOTS APPS FIGURE FORMAT TOOLS
MATLAB Drive
Files
DIP_1
Kuldip.jpg
lenna.m
chap2_1.m
chap2_2.m
chap2_3.m
chap5_1.m
chap5_2.m
chap5_3.m
chap8_1.m
Workspace
Name Value Size Class
gray_img2 1002×1000×3 ui... 1002×1000×3 uint8
img_color 1002×1000×3 ui... 1002×1000×3 uint8
img_gray 1002×1000 uint8 1002×1000 uint8
img_norm 1002×1000 dou... 1002×1000 double
lenna.m
/MATLAB Drive/DIP_1/lenna.m
clc;
clear;
close all;
% Read color image
img_color = imread('Kuldip.jpg');
% Display color image
figure, imshow(img_color), title('Color
% Convert to grayscale
img_gray = rgb2gray(img_color);
% Display grayscale image
figure, imshow(img_gray), title('Graysc
% Read grayscale image (optional)
gray_img2 = imread('Kuldip.jpg'); % rep
Command Window
Figure 1
Color Image
Figure 2
Grayscale Image
Figure 3
Normalized Image

Procedure:

1. Start MATLAB and open a new script file.
2. Clear previous data using clc, clear, and close all.
3. Load a color image using imread().
4. Display the image using imshow().
5. Convert the image into grayscale using rgb2gray().
6. Display the grayscale image.
7. Normalize pixel values by dividing by 255 after converting to double.
8. Display the normalized image.

Result:

- The original color image was successfully displayed.
- The grayscale image showed intensity variations without color.
- The normalized image displayed pixel values scaled between 0 and 1.

Discussion:

- Grayscale conversion reduces computational complexity.
- Normalization is essential for algorithms that require uniform data ranges.
- The double() function is necessary before normalization to avoid integer division.

Conclusion:

This experiment demonstrates basic image processing techniques in MATLAB, including reading, displaying, converting, and normalizing images. These steps form the foundation for advanced image processing tasks.